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Vishal Chand

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EDUCATION

IIT ROORKEE
INTEGRATED DUAL DEGREE (B.TECH + M.TECH)
2010 - 2015 | Roorkee, India
Cum. GPA: 8.677 / 10.0

LINKS

Personal:// **vishalchand.com**
Github:// **vish-chan**
LinkedIn:// **vishal-chand**

SKILLS

LANGUAGES

Expert: Java (runtime/tooling) • Python
Proficient: C++ (systems/runtime) • C
Exposure: Go • Rust • JavaScript • Android

SYSTEMS & RUNTIMES

OpenJDK • Java Virtual Machine (JVM) Internals • Hotspot VM
GC Design • Module System • Bytecode (ASM)

SYSTEMS & KERNEL

Linux Kernel Internals • Scheduler (CFS) • Memory Management
Virtual Memory • NUMA • Perf Events / PMU
Device Drivers • Android Kernel

PERFORMANCE ENGINEERING

Benchmarking (JMH, SpecJBB)
Profiling (JFR, async-profiler, perf)
Performance Modeling • Bottleneck Analysis

CLOUD & INFRASTRUCTURE

AWS (EC2, S3, EKS) • Cost Optimization
Distributed Systems • Kubernetes
Load Testing • Capacity Planning

TOOLS & FRAMEWORKS

Perf • Systrace • GDB • Git
AMD uProf • Scikit-learn

EXPERIENCE

AMAZON | SENIOR SOFTWARE DEVELOPMENT ENGINEER

Languages and Runtime | 2022 – Present | Dublin, Ireland

- **Metacircular Runtime Systems:** Designed and evolved a **metacircular Java runtime** atop JVM internals, delivering core runtime features, sustained execution-efficiency improvements, and deep fixes across classloading, reflection, and bytecode execution over multiple release cycles.
- **Performance & JVM Internals:** Reduced **module-loader latency by 96.8% (120μs ⍰ 3μs)**, unblocking Athena's JDK 11.17 adoption; eliminated Reflection API bottlenecks for a **40% Jython benchmark improvement**.
- **Systems Innovation:** Prototyped a **zero-overhead, agentless code coverage system** using JVM profiling data, enabling production-safe coverage without instrumentation or runtime overhead.
- **Distributed Systems Scaling:** Validated Pyroscope V2 horizontal scalability (1,700 profiles/s) on a 74-node cluster; removed Raft consensus and backend concurrency bottlenecks to achieve linear throughput scaling while reducing infrastructure footprint by **20–36%** under identical SLAs.
- **JDK Migration Tooling:** Built an automated bytecode- and API-level analyzer to accelerate JDK upgrades, reducing development cycles by **35%**.
- **Benchmark Infrastructure:** Built EC2-backed, fully automated benchmarking pipelines with provisioning and analysis, cutting manual performance evaluation effort by **80%**.
- **Mentorship & Leadership:** Mentored two interns on high-complexity runtime projects, leading to production deployment of a JVM flags translation feature and one full-time offer.

AMD | SENIOR SOFTWARE SYSTEM DESIGNER

Java Runtime & Performance | 2020 – 2022 | Bengaluru, India

- **HotSpot GC (C++):** Implemented **Configurable Card Size** (JDK-8272773) for G1, Parallel, and Serial GCs in **HotSpot C++**, improving remembered-set efficiency and delivering a **2% SPECjbb2015 gain** on large-heap workloads.
- **CPU-Aware JVM Performance:** Characterized and tuned **SPECjbb2015, SPECpower, Renaissance** on AMD EPYC, correlating JVM and GC behavior with cache, NUMA, and PMU data to identify VM optimization opportunities.
- **Benchmark Correctness (BenchSEE):** Identified and reported **incorrect CPU power benchmark implementations** in China's BenchSEE suite, driving upstream fixes and earning the **DSG Excellence Award**.
- **AutoTuner:** Built a black-box JVM auto-tuning framework for flags and workload parameters, cutting tuning cycles from **14 days to 3 days** with **99%+ accuracy**.
- **Ecosystem & Mentorship:** Contributed to Amazon Corretto's **Heapothesys** GC benchmark and mentored a postgraduate intern, reducing AutoTuner search time by **50%**.

ACHIEVEMENTS

2022	DSG Excellence (AMD)
2021	Spotlight (AMD)
2013-17	Qualstar (Qualcomm)
2014	Winner, IdeaQuest

GRADUATE COURSEWORK

- Advanced Algorithms
- Advanced Operating Systems
- Advanced Computer Architecture
- Advanced DBMS
- Machine Learning (Stanford)

TECHNICAL EXPERTISE

- **Runtime Systems:** OpenJDK, Hotspot VM, GC algorithms, bytecode & classloading
- **OS:** Linux kernel, syscalls, PMU, NUMA
- **Memory Systems:** GC design, virtual memory, page cache, huge pages
- **CPU Architecture:** Caches, TLBs, PMU counters, SMT, NUMA effects
- **Performance Engineering:** Profiling, benchmarking, root cause analysis
- **Distributed Systems:** Scalability analysis, Raft consensus
- **Cloud Infrastructure:** AWS Cloud, cost optimization, FinOps
- **Observability:** async-profiler, JFR, perf, Grafana Pyroscope

EXPERIENCE (CONTD.)

EXPERITEST | Software Engineer

Seetest Android | 2018 – 2019 | Delhi, India

- **Mobile Systems Engineering:** Developed core subsystems for Seetest Manual, a large-scale Android automation platform, improving reliability and execution efficiency of device orchestration and test workflows.
- **Cloud POCs:** Conducted architectural POCs and competitive analysis for cloud-based mobile testing infrastructure, informing scalability and deployment strategy.

QUALCOMM | Software Engineer

Android Performance | 2015 – 2018 | Hyderabad, India

- **SoC Performance Engineering:** Owned performance analysis for Qualcomm's **Snapdragon Wear** platform, optimizing boot time, CPU scheduling, and power-performance tradeoffs for resource-constrained devices.
- **Boot-Time Optimization:** Built a Python-based boot analyzer using USE methodology, enabling parallelized package scanning and reducing Android boot time by **10%**.
- **ML-Guided CPU Scheduling:** Designed an **ML-based CPU workload classifier** to dynamically select power-performance states, improving responsiveness under mixed workloads.
- **Graphics & Power Systems:** Diagnosed frame-drop issues in Wear OS via Systrace analysis and prototyped **suspend-to-disk** support for ultra-low-power wearable devices.

NOTABLE OPEN SOURCE CONTRIBUTIONS

OPENJDK | Contributor

- **JDK-8272773:** Implemented configurable card table sizing (128B-1024B) for G1, Parallel, and Serial GCs via `GCCardSizeInBytes` flag, optimizing Remembered Set overhead for large heaps.
- **JDK-8307732:** Backported build-test-lib infrastructure to JDK 17u, enabling WhiteBox testing library for jtreg test suite execution.

PERFORMANCE TOOLING | Community Contributor

- **async-profiler:** Implemented PMU counter scaling for kernel time multiplexing, improving concurrent profiler measurement accuracy.
- **Heapothesys:** Fixed survivor queue ordering and integer overflow for allocations $\geq$1024MB in Amazon Corretto's GC benchmark tool.